武汉大学

基于大语言模型的
端到端论辩挖掘方法研究

End-to-End Argument Mining
with Large Language Models

0914-v3

2026 9 14

0914-v2

2026 AAEC/CDCP

0914_v3

Single-Pass 81.67 77.82 Full 83.71 run ID A

14

MCR-BR

AAEC Component-F1 Relation-F1 83.71 64.83 CDCP 76.45 38.17 AAEC CDCP Link-F1 43.21 Single-Pass 9.79 F1

Abstract

Argument mining reconstructs components and relations from unstructured text under corpus-specific annotation guidelines. Generative large language models can represent component spans, types, and directed relations jointly. However, component errors restrict relation prediction, while a single generation may contain invalid references or unsupported links. This thesis studies complementary training-time conclusion enhancement and inference-time graph repair for end-to-end argument mining.

For training-time enhancement, this thesis designs local structure-preserving augmentation centered on conclusion components. A conclusion and its incoming neighbors define the editable nodes, and their annotated internal relations condition paraphrasing. Rewritten components are inserted into the original document with updated spans while preserving types, identifiers, and relation topology, followed by layered filtering. Type-conditioned low-rank adaptation, cue-aware continuous prompts, and a sentence-level auxiliary objective support representation learning. Whole-document reconstruction also duplicates unchanged components, so augmentation does not necessarily improve class balance.

For inference-time repair, this thesis proposes MCR-BR. Multiple component hypotheses receive relation predictions to form complete candidate graphs. Argument-link coverage, edge-level semantic coherence, and output validity guide initial selection; detected anomalies trigger type changes, component replacement, edge removal, or relation regeneration over specified source nodes. Edits are accepted only after validity and surrogate-score checks; otherwise, they are rolled back. Complete candidates contain both components and relations, but local repair cannot generally recover missing nodes. Constructed counterexamples clarify why higher surrogate scores need not indicate more accurate extraction.

The retained records report full-framework Component-F1/Relation-F1 scores of 83.71/64.83 on AAEC and 76.45/38.17 on CDCP. Under AAEC-to-CDCP zero-shot transfer with coarse component labels and directed unlabeled links, Link-F1 is 43.21, 9.79 points above the same-backbone Single-Pass baseline. These records indicate the potential value of combining enhancement and repair. Independent gains, matched-budget advantages, and broader transferability require configuration verification and corresponding ablations.

Keywords: Argument Mining; Large Language Models; Local Structure-Preserving Augmentation; Conclusion Components; Candidate-Graph Evaluation; Anomaly-Driven Repair

A

B

C

D

1.1 研究背景与问题提出

[1]

AAEC MajorClaim Claim Premise CDCP Fact Testimony Value Policy Reference [2] [3]

Claim

Claim

1.2 国内外研究现状

1.2.1 组件识别与联合结构预测

Stab Gurevych AAEC [2] Eger [4] Morio [5]

Caputo 2026 [6]

F1

1.2.2 生成式论辩挖掘与结构化输出

Kawarada [7] Cabessa PE AbstRCT CDCP

[8] ACC

Das AASP

[9] 2025 arXiv AASP “ ”

2026 ARGUS [10] AAAI

Li LLM

[11] “ ”

1.2.3 篇章结构、语义线索与关系推理

Sun DENIM [12]

2025 Das

[13] Sun FOL-AM

[14] F1

FOL-AM

1.2.4 结构保持数据增强

[15] CoDa

[16] SegMix

[17]

Claim

Claim Premise

2026 Sun LLM [18]

IoU

1.2.5 约束解码、多候选评估与局部修正

token [19] JSON

AASP

Self-Refine [20] Huang
[21] [22]

Tran LLM-refine Claim Evidence Warrant
LLM [23] “
”

Xiao Li 2026 ARGUS LLM
[24] Caputo AAAI
MCR-BR

1.2.6 中文资源、跨语料迁移与研究定位

Morio MT-AM [25]
“ ”

Savigny Yun 2026 AMELIA 19
[26]

Feger 2025 17 [27]
MCR-BR Cue-
aware

ORCHID AAEC
[28] IAM [29] AAEC CDCP

“ ”

1.3 研究问题与主要研究工作

1.3.1 研究问题

RQ1 Premise Precision Recall F1

RQ2

RQ3 AAEC CDCP

1.3.2 主要研究工作与贡献

1

2 MCR-BR

BR

3 AAEC CDCP Oracle

1-1

研究问题	主要方法	现有材料	尚缺的关键证据
RQ1			Premise
RQ2			
RQ3		AAEC CDCP	

1.4 论文组织结构

–

MCR-BR

2.1 论辩组件与关系表示

$$D=(x_1,\dots,x_n) \qquad c_i=(s_i,e_i,\tau_i) \qquad 1\le s_i\le e_i\le n \qquad (1)$$

τ_i token token

$C=\{c_1,\dots,c_m\}$ (i,j,r) i j

$G=(V,E)$ “ ”

2-1

语料	组件类型	本文域内关系标签	需要区分的属性或结构
AAEC	MajorClaim Claim Premise	Support Attack	Claim MajorClaim stance F1
CDCP	Fact Testimony Value Policy Reference	Reason Evidence	

2-1 [2][3][7] AAEC Claim MajorClaim Support Attack MajorClaim CDCP Claim/Premise

2.2 合法图与序列化约定

$\mathcal{A}$ $\mathcal{G}_{\mathcal{A}}$

“ ” 0 0 0

ID token “Libraries / should / stay / open / longer / . / Students / need / study / space / .” Claim 1 5 Premise 7 10

```
{
  "components": [
    {"id": 1, "type": "Claim", "span": [1, 5]},
    {"id": 2, "type": "Premise", "span": [7, 10]}
  ],
  "relations": [
    {"source": 2, "target": 1, "type": "Support"}
  ]
}
```

2.3 生成建模与参数高效适配

$$Y=(y_1,\ldots,y_L)$$

$$P_\theta(Y\mid D)=\prod_{t=1}^{L}P_\theta(y_t\mid D,y_{<t}). \quad 2\text{-}1$$

Transformer [30] Qwen2.5-7B-Instruct Qwen2.5 [31]

LoRA [32]

$$W=W_0+\frac{\alpha}{r}BA,\ A\in\mathbb{R}^{r\times d_{in}},\ B\in\mathbb{R}^{d_{out}\times r}. \quad 2\text{-}2$$

r α LoRA [33] LoRA

2.4 候选选择与代理评分

$$\mathcal{S}=\{S_1,\ldots,S_K\}$$

$$S^*=arg\,max_{S\in\mathcal{S}}\,E(S). \quad 2\text{-}3$$

Candidate Oracle

2.5 评价指标与可比性条件

TP FP FN F1

$$P=\frac{TP}{TP+FP}, R=\frac{TP}{TP+FN}, F_1=\frac{2TP}{2TP+FP+FN}. \quad 2\text{-}4$$

micro-F1 F1 macro-F1 macro-F1

micro/macro “±”

F1 “F1 ”

0914-v2 ID FP

FN ID 0

“±”

Link-F1 Relation-F1 F1

2.6 训练增强与推理修正的接口

Full

2.7 本章小结

Claim

Claim-specific Adapter

Cue-aware Soft Prompting

MajorClaim Claim

3.1 组件不平衡与结构影响

3.1.1 统计口径

$\mathcal{T}$ τ N_τ

$$IR=\frac{max_{\tau\in\mathcal{T}}\ N_\tau}{min_{\tau\in\mathcal{T}}\ N_\tau}. \quad 3\text{-}1$$

token BIO

R01 IR “ ” “ ”

MajorClaim Claim Premise

AAEC Claim MajorClaim stance Claim

MajorClaim “ ”

3.1.2 错误传播路径

C $\hat{C}$ $\hat{C}\times\hat{C}$ V_{hit}

$$E_{reach}=\{(i,j,r)\in E: i\in V_{hit}, j\in V_{hit}\}. \quad 3\text{-}2$$

$|E|>0$

$$R_{rel}\le\frac{|E_{reach}|}{|E|}. \quad 3\text{-}3$$

E_{reach}

MajorClaim

Claim Premise

3.2 结论中心的局部结构保持增强

3.2.1 局部子图作为增强单位

$G=(V,E)$ v_c

$$V_c=\{v_c\}\cup\{v_i/(v_i,v_c,r)\in E\},G_c=(V_c,E_c).$$

$$E_c=\{(i,j,r)\in E:i\in V_c,j\in V_c\}.$$

AAEC Claim—MajorClaim Support/Attack MajorClaim ID

3.2.2 条件生成约束

G_c ID

$$D_c'=LLM_{aug}(D_c/\mathcal{T}_c,E_c,\Pi), \quad 3\text{-}4$$

D_c $\mathcal{T}_c$ E_c Π

ID

3.2.3 三层质量筛选

(D_c',G_c')

1. 格式合法性 JSON
2. 结构一致性
3. 语义一致性

q_{fmt} q_{str} q_{sem}

$$q(D_c', G_c') = q_{fmt} \cdot q_{str} \cdot q_{sem} = 1. \quad 3\text{-}5$$

3.2.4 增强效果的可识别性与数据追踪

Claim k Premise u

$N_C + u$ $N_P + k u$ $u > 0$ Claim/Premise $N_P > k N_C$

d τ $N_{d,\tau}$ a_d

$N_\tau' = N_\tau + \sum_d a_d N_{d,\tau}.$

IR

MajorClaim stance AAEC

R02

ID ID

ID 4 2 1

plain

3.2.5 全文回填的构造示例

“Libraries should stay open longer. Students need study space. Campus services need investment.” 2.1 1 Claim [1,5] 2 Premise [7,10] 3 Claim [12,15] Support $2 \to 1$ $2 \to 3$

1 1 2 $2 \to 1$ $2 \to 3$ “Libraries ought to remain open longer” “Students require space for studying” 3 [1,6] [8,12] [14,17] $2 \to 1$ $2 \to 3$ 3

ID 2 “ ”

3.3 □□□□□□□□□□□□□

DENIM [12]

[13]

3.3.1 Claim-specific Adapter

0914-v2 g c W_0

h_t

$$h_t' = W_0 h_t + \frac{\alpha}{r} B_g A_g h_t + g_t \frac{\alpha}{r} B_c A_c h_t, \quad 3\text{-}6$$

d $r=16$ $\alpha=32$ $g_t \in \{0,1\}$

JSON

R03 token Claim span type

id type span token

AAEC Claim/MajorClaim CDCP Policy/Value

LoRA LoRA rank32 rank16 LoRA

“ ”

span F1

3.3.2 Cue-aware Soft Prompting

“therefore”“thus”“in conclusion” $m=4$

$$X' = [P_1, P_2, \ldots, P_m; X], \quad 3\text{-}7$$

$P_i \in \mathbb{R}^d$

therefore thus in conclusion consequently hence to conclude

4

3.3.3 多目标训练

token EOS token

T_y

$$\mathcal{L}_{gen}=-\frac{1}{T_y}\sum_{t=1}^{T_y}\log P\left(y_t \mid D, y_{<t}\right). \quad 3\text{-}8$$

token N_s

$z_i=1$ 0 AAEC MajorClaim/Claim CDCP Policy/Value

$$\mathcal{L}_{sat}=-\frac{1}{N_s}\sum_{i=1}^{N_s}\left[z_i\log\hat{z}_i+\left(1-z_i\right)\log\left(1-\hat{z}_i\right)\right]. \quad 3\text{-}9$$

$$\mathcal{L}_{comp}=\mathcal{L}_{gen}+\lambda\mathcal{L}_{sat},\lambda=0.5. \quad 3\text{-}10$$

JSON $\lambda=0.5$

R04 loss mask

3.4 实验设置与结果分析

3.4.1 设置与评价指标

Qwen2.5-7B-Instruct LoRA F1 Component-F1 MajorClaim-F1 Claim-F1

3-1 Single-Pass Comp-F1 81.67 77.82 Full 83.71 Full run ID

R05

3.4.2 主要结果

3-1 AAEC

方法	MajorClaim-F1	Claim-F1	Comp-F1
FLAN-T5 (XXL)	71.23	76.45	80.15

方法	MajorClaim-F1	Claim-F1	Comp-F1
LLM Single-Pass	73.45	78.21	81.67
Ours (Full)	**77.34**	**81.77**	**83.71**

Full Single-Pass MajorClaim-F1 Claim-F1 Component-F1 3.89 3.56 2.04 F1 FLAN-T5 (XXL) 3.56 Premise Precision Recall F1 F1

3.4.3 模块作用与解释边界

Claim-specific Adapter Component-F1 1.75 Cue-aware Soft Prompting 1.37 0.82 “ ”

3.5 误差类型与讨论

Cue-aware MajorClaim Claim

3.6 本章小结

Premise

MCR-BR

“ ”

4.1 单次生成的错误传播与恢复目标

MCR-BR

AAEC 42.5% 18.5% JSON 6.2% 22.4%

BR

Claim

4.2 多候选组件生成与完整图构建

4.2.1 候选组件集合

D K

$$\mathcal{C}=\{C^{(1)},C^{(2)},\dots,C^{(K)}\}. \quad 4\text{-}1$$

$C^{(k)}$ JSON Schema

R06

ID

4.2.2 候选论辩图构建

$C^{(k)}$ $R^{(k)}$

$$S_k=(C^{(k)},R^{(k)},G^{(k)}),G^{(k)}=(V^{(k)},E^{(k)}). \quad 4\text{-}2$$

$C^{(k)}$ ID

ID ID AAEC Support/Attack CDCP Reason/Evidence Link

4-1 MCR-BR

↓
K
↓
K
↓

↓
→ →

JSON BR

JSON 6.2% 0.0%

4.3 一致性引导的候选评分

$$E(S_k)=\alpha E_{dens}(S_k)+\beta E_{coh}(S_k)+\gamma E_{val}(S_k), \quad 4\text{-}3$$

$\alpha=0.4$ $\beta=0.4$ $\gamma=0.2$ $[0,1]$

4.3.1 论证连接覆盖

$\mathcal{Q}(S_k)$ AAEC MajorClaim

(i,j,r_{ij}) $p_{ij}\in\{0,1\}$

v_j

$$d_{coh}^{in}(v_j)=\sum_{(i,j,r_{ij})\in R^{(k)}} p_{ij}. \quad 4\text{-}4$$

$$E_{dens}(S_k)=\begin{cases}\dfrac{1}{|\mathcal{Q}(S_k)|}\sum_{v_j\in\mathcal{Q}(S_k)} min\left(\dfrac{d_{coh}^{in}(v_j)}{\tau(t_j)},1\right), & |\mathcal{Q}(S_k)|>0,\\ 0, & |\mathcal{Q}(S_k)|=0.\end{cases} \quad 4\text{-}5$$

$\tau(t_j)$ t_j τ_j 2 1

Attack “ ” “ ”

Claim

$\mathcal{Q}$

F1

AAEC Claim CDCP Policy/Value 2

1 0

4.3.2 语义连贯性

$$E_{coh}(S_k)=\begin{cases}\dfrac{1}{|R^{(k)}|}\sum_{(i,j,r_{ij})\in R^{(k)}} p_{ij}, & |R^{(k)}|>0,\\ 0, & |R^{(k)}|=0.\end{cases} \quad 4\text{-}6$$

E_{coh}

p_{ij}

F1

4.3.3 输出合法性

$E_{val}(S_k)=1$ 0 JSON token

Schema

$$k^{*}=\arg\max_{1\le k\le K} E(S_k), S^{(0)}=S_{k^{*}}. \quad 4\text{-}7$$

4.4 图约束下的有界局部修正

4.4.1 异常表示与优先级

S

$$\mathscr{E}(S)=\{e=(v,\eta,\sigma)\}, \quad 4\text{-}8$$

$v\in\{VAL,COH,DENS\}$ η $\sigma\in[0,1]$

$$\sigma(e)=\begin{cases}1, & v=VAL,\\ 1-p_{ij}, & v=COH,\eta=(i,j),\\ 1-\min\left(\dfrac{d_{coh}^{in}(v_j)}{\tau(t_j)},1\right), & v=DENS,\eta=j.\end{cases} \quad 4\text{-}9$$

$$VAL \succ COH \succ DENS$$ “

$\delta_{dens}=1.0$” $min(d_{coh}^{in}/\tau,1)$ 1 R07

4.4.2 修复动作

BR

1. **Promote/Demote** MajorClaim Claim

2. **Replace**

3. **RemoveEdge**

4. **LocalRerun**

a $a(S)$

$$Accept(a,S)=\mathbb{I}[a(S)\in\mathcal{G}_{\mathcal{A}}]\cdot\mathbb{I}[E(a(S))\geq E(S)]. \quad 4\text{-}10$$

Replace

Claim LocalRerun Promote/Demote

AAEC stance R08 ID

4-1

异常对象	候选动作与顺序	字段范围与执行限制
VAL COH	RemoveEdge LocalRerun	
VAL	Replace LocalRerun	AAEC Claim/MajorClaim Replace ID
DENS	Replace LocalRerun	
JSON		BR

4-1 Replace

ID ID

LocalRerun

DENS “ ”

4 LocalRerun

4.4.3 终止条件与计算预算

$T=3$ $M=2$ GenerateActions

4-1

```
        {S1,...,SK}     T       M
          S

S ← argmax E(Sk)
for t = 1,...,T:
    errors ← DetectErrors(S)
    if errors       :
        break
    accepted ← 0
    for e in SortBySeverity(errors):
        for a in GenerateActions(e, candidate_pool):
            S' ← a(S)
            if Valid(S') and E(S') ≥ E(S):
                S ← S'
                accepted ← accepted + 1
                break
        if accepted = M:
            break
    if accepted = 0:
        break
return S
```

$TM=6$ 6

24 4 128 32768

token 1024 32

4.4.4 代理目标与可恢复性分析

$$E(S')\geq E(S)$$

Relation-F1

1 0.8 0.5 0.6

ID BR

Claim Premise Claim 1
Premise Support Claim 2 Claim 2 0.5 1
0.8 Claim 1 MajorClaim 1 1
DENS

F1 100 75 F1 100

B.5

“ ” “ ” B

4.5 □□□□

4.5.1 □□□□□□

AAEC CDCP F1

$$P=\frac{TP}{TP+FP}, R=\frac{TP}{TP+FN}, F_1=\frac{2PR}{P+R}. \quad 4\text{-}11$$

TP TP AAEC
Component-F1 Support Attack Relation-F1 CDCP Reason Evidence
F1 Oracle

4.5.2 □□□□□

FLAN T5-XXL GPT-4-turbo 20-shot RoBERTa-large
Qwen2.5-7B-Instruct LoRA Single-Pass Constrained Decoding DAG
Post-process Multi-Sample (LogProb) Self-Refine $K=1$ $K=3$
AdamW [7]

R09 ID CDCP batch size
LoRA dropout top-p “±”

R10 FLAN T5-XXL AAEC 80.15/61.19 CDCP 72.68/33.96 GPT-4-turbo
AAEC 55.51/28.38 Kawarada RoBERTa
Self-Refine MCR-BR token

4.5.3 近期相关方法的比较边界

DENIM[12] AASP[9]

Cabessa [8] FOL-AM

Tran LLM-refine[23]

ARGUS

4.5.4 配置核对与新实验命名

LoRA K/T/M “ ”

“ ” ID dropout D

3-1 5-2

Single-Pass staged-K1 K3-LogProb

K3-Rerank K3-Rerank-BR

81.67 77.82

3 epoch /token

4.6 域内主要结果

4.6.1 AAEC 结果

4-2 AAEC

方法	Comp-F1	Support-F1	Attack-F1	Relation-F1
FLAN T5-XXL	80.15	62.34	48.72	61.19
GPT-4-turbo 20-shot	55.51	29.86	18.45	28.38
RoBERTa-large MTL	76.43±0.55	54.21±0.52	41.83±0.67	52.15±0.54
LLM Single-Pass	77.82±0.51	58.43±0.45	44.21±0.62	56.87±0.48
LLM w/ Constrained Decoding	78.54±0.46	59.12±0.42	45.33±0.58	57.65±0.45
LLM w/ DAG Post-process	78.91±0.49	59.45±0.48	45.87±0.61	58.12±0.47
LLM Multi-Sample (LogProb)	79.45±0.38	60.12±0.41	46.58±0.55	58.94±0.42
LLM w/ Self-Refine	79.82±0.45	60.75±0.43	47.12±0.56	59.63±0.45
Ours ($K=1$, w/o correction)	81.24±0.44	61.87±0.39	48.34±0.58	60.73±0.41

□□	Comp-F1	Support-F1	Attack-F1	Relation-F1
Ours (Verifier-only Rerank, $K=3$)	82.18±0.40	63.15±0.36	49.82±0.52	62.21±0.38
Ours (Full, $K=3$)	**83.71±0.42**	**65.92±0.38**	**51.47±0.54**	**64.83±0.38**

4-2 Full FLAN T5-XXL Component-F1 Relation-F1 3.56 3.64 Single-Pass 5.89 7.96 DAG Self-Refine BR

4.6.2 CDCP □□

4-3 CDCP

□□	Comp-F1	Reason-F1	Evidence-F1	Relation-F1
FLAN T5-XXL	72.68	35.24	28.41	33.96
RoBERTa-large MTL	68.52±0.58	31.45±0.61	25.72±0.73	30.23±0.64
LLM Single-Pass	71.35±0.48	33.82±0.52	27.15±0.68	32.64±0.55
LLM w/ DAG Post-process	72.14±0.50	34.65±0.49	28.02±0.63	33.51±0.52
LLM w/ Self-Refine	72.95±0.47	35.58±0.48	28.94±0.60	34.42±0.50
Ours ($K=1$, w/o correction)	74.56±0.45	36.94±0.46	30.27±0.58	35.78±0.48
Ours (Verifier-only Rerank, $K=3$)	75.12±0.41	37.85±0.44	31.42±0.55	36.64±0.45
Ours (Full, $K=3$)	**76.45±0.43**	**39.12±0.45**	**32.86±0.57**	**38.17±0.46**

4-3 CDCP Full Relation-F1 FLAN T5-XXL 4.21 AAEC Evidence-F1 Reason-F1

4.7 □□□□□□□

4.7.1 □□□□

4-4

□□	Comp-F1	Relation-F1	ΔComp	ΔRel
Full Method	83.71±0.42	64.83±0.38	—	—
$K=1$, w/o correction	81.24±0.44	60.73±0.41	-2.47	-4.10
w/o Claim-specific Adapter	81.96±0.48	63.08±0.44	-1.75	-1.75

配置	Comp-F1	Relation-F1	ΔComp	ΔRel
w/o Cue-aware Prompting	82.34±0.45	62.87±0.40	-1.37	-1.96
w/o Multi-Objective	82.89±0.43	63.54±0.39	-0.82	-1.29
w/o Validity Signal E_{val}	81.56±0.46	61.92±0.43	-2.15	-2.91

$K=1$ $K=3$ Relation-F1 1.48

$K=3$ Full 2.62 BR

E_{val} 2.91 Relation-F1

E_{dens} E_{coh}

—

4.7.2 验证器替换分析

4-5

验证器	Comp-F1	Relation-F1	Top-1 一致率	修正触发率
Qwen2.5-7B	83.71	64.83	100.0%	42.5%
Llama-3-8B	83.71	64.55	94.2%	43.8%

Relation-F1 -0.28 Top-1 94.2% 100.0%

R11

Precision Recall

4.7.3 Oracle 上限

4-6 Oracle

条件	Comp-F1	Relation-F1	ΔRel
	83.71±0.42	64.83±0.38	—
Candidate Oracle $K=3$	85.12±0.36	67.24±0.34	+2.41
Condition Oracle	—	78.42±0.35	+13.59
Output Oracle	100.00	83.35±0.32	+18.52

4-6 Candidate Oracle Relation-F1 2.41 Condition Oracle 13.59 Oracle

13.59

R12 Candidate Oracle Output Oracle

Relation-F1 100 83.35

4.7.4 修正行为统计

4-7

指标	结果
	42.5%
	1.8
	2.4
LocalRerun	45.2%
RemoveEdge	24.8%
Replace	19.5%
Promote/Demote	10.5%
ID	18.5% → 2.1%
JSON	6.2% → 0.0%
	22.4% → 5.8%
E_{val}	75.3% → 97.9%

LocalRerun

RemoveEdge Precision Recall

R13 ID

75.3% 97.9% 6.2% JSON 18.5%

ID

4.8 误差分析与局限

“ Claim” Replace

Claim Premise

Support Attack 1 0 0.5

0.5 1 0.6 COH RemoveEdge 0.5 1 1 0.8

F1 100 F1 66.67 100 DENS

MCR-BR $K=3$ $T=3$ $M=2$

A_{try}

token P95 Single-Pass

GPU

4.9 □□□□

Oracle BR

MT-AM[25] [27]

5.1 跨领域异构性

AAEC CDCP AAEC CDCP
AAEC MajorClaim Claim Premise CDCP Fact Testimony
Value Policy Reference AAEC Support/Attack Claim stance CDCP
Reason/Evidence F1 F1

5.2 粗粒度标签与关系映射

$$\varphi: \mathcal{T}_{src} \cup \mathcal{T}_{tgt} \to \{Claim, Premise\}$$

5-1

数据集原标签	粗粒度标签	映射依据
AAEC MajorClaim	Claim	
AAEC Claim	Claim	
AAEC Premise	Premise	
CDCP Policy	Claim	
CDCP Value	Claim	
CDCP Fact	Premise	
CDCP Testimony	Premise	
CDCP Reference	Premise	

ψ AAEC Support/Attack CDCP Reason/Evidence Link

$$\psi(r) = Link, r \neq None. \quad 5\text{-}1$$

/ / Unlabeled Link-F1

“ ”

5.3 零样本迁移协议

AAEC AAEC CDCP

AAEC CDCP Link

Schema

1. **Coarse Component-F1** Claim/Premise

2. **Unlabeled Link-F1** Reason Evidence

Link Schema

R14

5.4 跨领域迁移结果

5-2 AAEC CDCP

方法	**Coarse Component-F1**	**Unlabeled Link-F1**
FLAN T5-XXL zero-shot	52.34±0.65	31.25±0.72
LLM Single-Pass zero-shot	54.87±0.58	33.42±0.68
Ours ($K=1$, w/o correction, zero-shot)	57.23±0.52	38.76±0.61
Ours (Full, zero-shot)	**60.78±0.48**	**43.21±0.55**

5-2 Full FLAN T5-XXL Coarse Component-F1 Unlabeled Link-F1 8.44 11.96 Single-Pass 5.91 9.79

43.21 Link-F1 AAEC 64.83 Relation-F1

R14 Link Schema

5.5 泛化来源与有效性威胁

1. AAEC CDCP

2. Policy/Value Claim Fact/Testimony/Reference Premise

3. Unlabeled Link-F1

4.

5.

Cue-aware

AbstRCT

5.6 □□□□

AAEC CDCP Full

6.1 工作总结

MCR-BR

Full AAEC Component-F1 Relation-F1
83.71 64.83 CDCP 76.45 38.17 AAEC CDCP
60.78 43.21

RQ1 Premise RQ2
RQ3

6.2 当前不足与后续研究

“±”

Premise

– –token

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A

14

A-1

编号	位置与事项	所需证据
R01	3.1	IR
R02	3.2	
R03	3.3	token
R04	3.3	mask
R05	3.4	81.67 77.82 83.71 run ID
R06	4.2	ID
R07	4.3 4.4	
R08	4.4	
R09	4.5	ID
R10	4.5	
R11	4.7	
R12	4.7 Oracle	
R13	4.7	
R14	5.3 5.4	

Single-Pass Full BR

B

/ /token

Insert

B.1 合法候选优先与失败处理

B.2 非法状态下的阶段性修复

“ ”

B.3 节点补回与总调用预算

Claim Insert ID

token 4.4.3

—

B.4 建议的可复现实验记录

4.4.4

ID token

F1

B.5 覆盖评分投机的保守修订方向

“ ” “ ” ID

Promote/Demote Replace

F1 F1 BR

B.6 两项贡献的最小独立验证

plain

BR token

BR 2×2 / / BR

Precision Recall F1 TP/FP/FN

micro-F1 plain BR

C

0914 0913 academic-research-suite

Word

ARS

“ ”

0914-v2 “ ”

0914

0914-v3 “ ” academic-research-suite

2026

ARS

D

D.1 身份与配置来源

argument_mining_0914 0914-reference-v1

19 D-1 “ ”

D-1

配置组	新实现取值或规则	来源及限制
	Qwen2.5-7B-Instruct r=16 alpha=32 m=4 0.5	
	0.4/0.4/0.2 K=3 T=3 M=2	M
	AdamW lr=0.0002 3 1 8 dev	
	q_proj/v_proj dropout=0.05 seed=17	
	temperature=0.7 top-p=0.9 1024 token 8192	
	CUDA/bfloat16 Torch 2.8.0 Transformers 4.57.6	GPU CPU
	24 128 32768 token	
	train/dev/test ID micro-F1	

Full train

D.2 输入输出与可追溯性

AAEC BRAT stance Claim MajorClaim CDCP Reason/ Evidence expand Unicode token

seed

D.3 当前验证范围与未完成工作

55

Qwen mask Qwen2.5-7B

validation

Self-Refine Llama Oracle Insert

GPU token “

”